CHEMISTRY

SECTION – I
Single Correct Choice Type

This section contains 6 single correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

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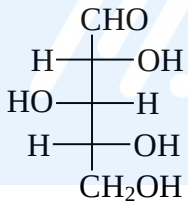
1. A 150 mL of solution of I_2 is divided into two unequal parts. 1st part reacts with hypo solution in acidic medium. 15 mL of 0.4 M hypo was consumed. 2nd part was added to 100 mL of 0.3 M NaOH solution. Residual base required 10 mL of 0.3 M H_2SO_4 solution for complete neutralization. What was the initial concentration of I_2?

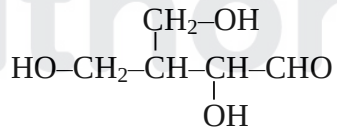
(a) 0.08 M (b) 0.1 M (c) 0.2 M (d) none of these

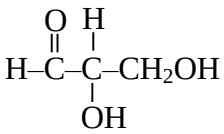
2. Metal 'M' + air $\xrightarrow{\Delta}$ (A) $\xrightarrow{H_2O}$ (B) $\xrightarrow{HCl}$ White fumes. Metal 'M' can be

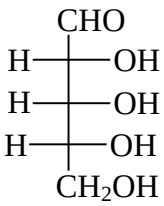
(a) Li, Mg or Al (b) Li, Al or K (c) Na, K or Mg (d) Li, Na or K

3. Compound (A) of molecular formula, $C_5H_{10}O_4$ is oxidised by Br_2/H_2O to, give acid (B) of molecular formula $C_5H_{10}O_5$ which readily forms a lactone on heating. (A) forms a triacetate with Ac_2O and an osazone with $PhNHNH_2$, (A) is oxidised by HIO_4. Only one mole of HIO_4 is consumed per mol of (A). The structure of (A) will be

(a)


(b)


(c)


(d)


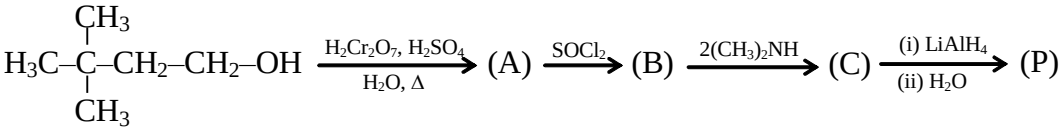
4. Consider the following reaction in aqueous solution at equilibrium.

$6HCHO(aq) \rightleftharpoons C_6H_{12}O_6(aq)$

What is the degree of association of HCHO in the above reaction if observed molar mass of HCHO and $C_6H_{12}O_6$ in the mixture is 150.

(a) 0.5 (b) 0.833 (c) 0.9 (d) 0.96

5. Identify product (P) in the following reaction sequence.



(a) $(CH_3)_3C–CH_2CN$

(b) $(CH_3)_3C–CH_2–CH_2N(CH_3)_2$

(c) $(CH_3)_3C–CH_2–\underset{\displaystyle N(CH_3)_2}{\underset{|}{CH}}–N(CH_3)_2$

(d) $(CH_3)_3C–CH_2–\underset{\displaystyle OH}{\underset{|}{CH}}–N(CH_3)_2$

6. Match List-I with List-II and select the correct answer

List-I	**List-II**
(A) $AlCl_3$ if $\propto = 0.8$	**1.** $i = 3.4$
(B) $BaCl_2$ if $\propto = 0.9$	**2.** $i = 2.8$
(C) Na_3PO_4 if $\propto = 0.9$	**3.** $i = 3.8$
(D) $K_4[Fe(CN)_6$ if $\propto = 0.7$	**4.** $i = 3.7$

	(A)	**(B)**	**(C)**	**(D)**
(a)	1	2	4	3
(b)	2	1	3	4
(c)	1	4	2	3
(d)	4	2	3	1

SECTION – II
Multiple Correct Choice Type

This section contains 5 multiple correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONE OR MORE** is/ are correct.

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7. The radial distribution function [P(r)] is used to determine the most probable radius which is used to find the electron in a given orbital. $\frac{dP(r)}{dr}$ for 1s-obtital of hydrogen like atom having atomic number 'Z' is

$$\frac{dP(r)}{dr} = \frac{4Z^3}{a_o^3}\left[2r - \frac{2Zr^2}{a_o}\right]e^{-2Zr/a_o}$$

then which of the following statements is/are correct?

(a) At the point of maximum value of radial distribution function $\frac{dP(r)}{dr}$ = zero, one antinode is present.

(b) Most probable radius of Li^{2+} is $\frac{a_o}{3}$ pm.

(c) Most probable radius of He^+ is $\frac{a_o}{2}$ pm.

(d) Most probable radius of hydrogen ion (H^+) is a_o pm.

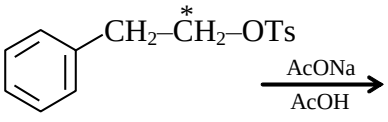
8. The possible product(s) formed in the following reaction is/are

$Cl_2O_6 + NaOH \longrightarrow ?$

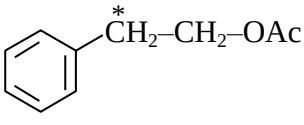
(a) $NaClO_4$ (b) $NaOCl$

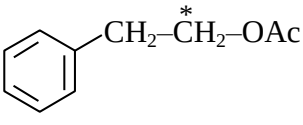
(c) $NaClO_2$ (d) $NaClO_3$

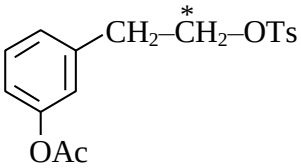
9. Product(s) formed during this reaction are

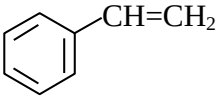


[C* = isotopic carbon]

(a) 

(b) 

(c) 

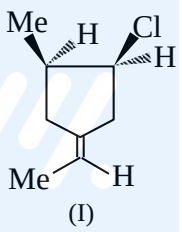
(d) 

10. In which of the following pair(s), the minerals are converted into metals by self reduction?

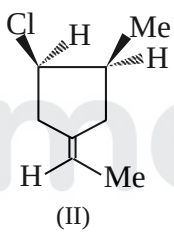
(a) Ag_2S, ZnS (b) Cu_2S, HgS

(c) PbS, ZnS (d) Cu_2S, PbS

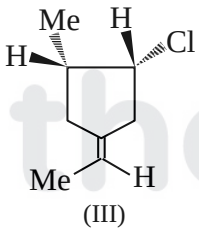
11. Consider the structures shown below:



(I)



(II)



(III)

and select the correct choice(s).

(a) (I) and (II) are geometrical isomer. (b) (I) and (III) are enantiomers.

(c) (II) and (III) are homomers. (d) (I) and (II) are enantiomers.

SECTION – III

Reasoning Type

This section contains 5 reasoning type questions. Each question contains Statement-1 and Statement-2. Each question has 4 choices (a), (b), (c) and (d), out of which **ONLY ONE** is correct.

Directions: Read the following questions and choose

(A) **Statement-1** is True, **Statement-2** is True; **Statement-2** is a correct explanation for **Statement-1**.

(B) **Statement-1** is True, **Statement-2** is True; **Statement-2** is not a correct explanation for **Statement-1**.

(C) **Statement-1** is True, **Statement-2** is False.

(D) **Statement-1** is False, **Statement-2** is True.

12. **Statement-1:** An optically active molecule must be asymmetric.

Statement-2: Asymmetric molecules will have non-superposable mirror image.

(a) A (b) B (c) C (d) D

13. **Statement-1:** For a real gas 'X', the Joule-Thomson coefficient has non-zero value.

Statement-2: The gas 'X' has intermolecular interactions.

(a) A (b) B (c) C (d) D

14. **Statement-1:** A certain reaction with $\Delta_r G° > 0$ will be non-spontaneous.

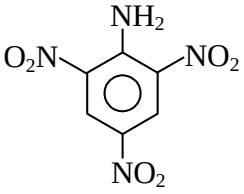
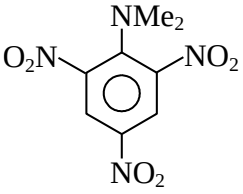
Statement-2: $\Delta_r G°$ can be used to predict the spontaneity of any reaction under standard conditions.

(a) A (b) B (c) C (d) D

15. **Statement-1:** The number of nodes in the radial function is equal to n-*l*-1.

Statement-2: The point at which r = 0, is not considered as node.

(a) A (b) B (c) C (d) D

16. **Statement-1:**  is less basic than  .

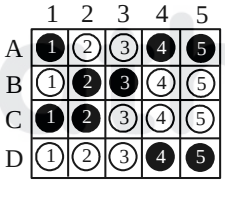
Statement-2: Only the +I effect of two methyl groups on nitrogen makes N,N-dimethylpicramide more basic than picramide.

(a) A (b) B (c) C (d) D

SECTION–V
Matrix–Match Type

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled 1, 2, 3, 4 and 5. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II**. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – 1, 4 and 5; B – 2 and 3; C – 1 and 2; and D – 4 and 5; then the correct darkening of bubbles will look like the following.

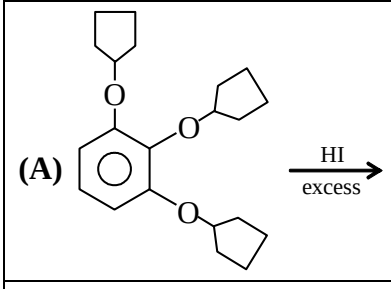
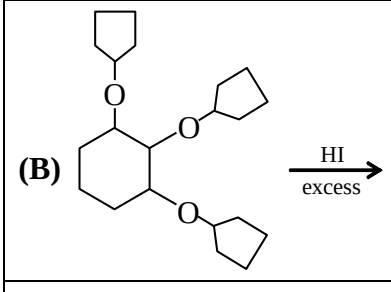
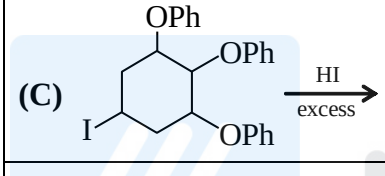
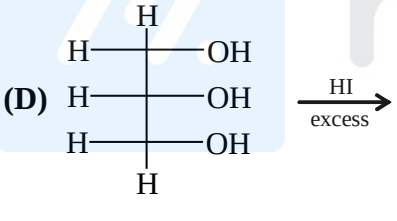
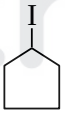


1.

Column I	Column II
(A) Reversible cooling of an ideal gas at constant volume.	**1.** $w = 0, q < 0, \Delta U < 0$
(B) Reversible isothermal expansion of an ideal gas.	**2.** $w = 0, q > 0, \Delta U > 0$
(C) Adiabatic expansion of non-ideal gas into vacuum.	**3.** $w = 0, q = 0, \Delta U = 0$
(D) Reversible melting of sulphur at normal melting point.	**4.** $w < 0, q > 0, \Delta U = 0$ **5.** $\Delta H \neq 0$

mathongo

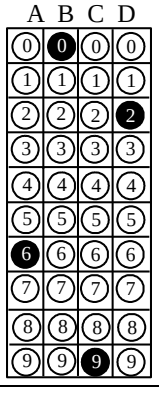
2.

Column I	Column II
(A) 	**1.** More than 4-moles of HI will be consumed.
(B) 	**2.** Product will react with Na.
(C) 	**3.** Product formed contains 2° halides.
(D) 	**4.**  is one of the product.
	5. One of the products contains more than one I-atoms.

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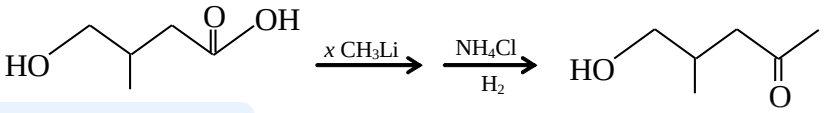
SECTION–V
Integer Answer Type

This section contains 4 questions. The answer to each of the questions is a **single digit integer,** ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers A, B, C and D (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the following:



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1. When carboxylic acid reacts with organolithium reagents to give ketones, side reactions sometime occurs, for example.



value of (x) in the above reaction is

Space for work

mathongo

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2. If the slope of 'Z' (compressibility factor) v/s 'P' curve is constant $\left(\text{slope} = \frac{\pi}{492.6}\text{atm}^{-1}\right)$ at a particular temperature (300 K) and very high pressure, then calculate diameter of the molecule (in Å) ($N_A = 6 \times 10^{23}$, R = .0821 atm L, $\text{mol}^{-1}\ \text{K}^{-1}$)

Space for work

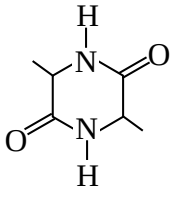
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3. A fresh radioactive mixture containing short lived species X and Y. Both the species together emitting 8000 α-particles per minute initially. 20 minutes later X was emitting α-particles at the rate of 4500 per minute. If the half-lives of the species X and Y are 10 minute and 500 hours, then what is the ratio of initial activities of X and Y in the mixture?

Space for work

mathongo

4. The number of stereoisomer formed by the given compound is



Space for work

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